

## Algebra 1: Unit 8, Lesson 8

1. Three different forms of a quadratic equation are shown.

| Standard Form       | Factored Form         | Vertex Form              |
|---------------------|-----------------------|--------------------------|
| $y = 2x^2 + 6x + 4$ | $y = 2(x + 1)(x + 2)$ | $y = 2(x + 1.5)^2 - 0.5$ |

The  $y$ -intercept(s) is/are \_\_\_\_\_ and can be determined by the standard form.

2. Three different forms of a quadratic equation are shown.

| Standard Form       | Factored Form          | Vertex Form                |
|---------------------|------------------------|----------------------------|
| $y = 4x^2 + 2x - 6$ | $y = 2(2x + 3)(x - 1)$ | $y = 4(x + 0.25)^2 - 6.25$ |

What is the vertex of the quadratic equation?

3. An equation is shown.

$$F(x) = -3x^2 + 6x - 5$$

Find  $F(4)$ .